

Olist Seller Performance Analysis

Jessica Leo

Identifying Key Success Factors of Sellers on Brazil's Largest E-Commerce Platform

Background & Problem Statement



Olist is Brazil's largest e-commerce marketplace connecting thousands of sellers with millions of customers. Understanding seller performance is critical to sustaining platform growth and preventing top sellers from switching to competitors.

Analysis Questions:

1. Which sellers are most valuable based on revenue and order volume?
2. How do revenue and order trends change over time?
3. Which regions produce the best-performing sellers?
4. Which product categories dominate platform sales?

Methodology

Tools: SQL (SQLite) + Microsoft Excel (Power Query, Pivot Table, Pivot Chart)

Key Variables: seller_id, revenue, order volume, state, product category, payment_value, order_status, order_purchase_timestamp

Tables: sellers, orders, order_items, products, customers, order_payments

Data Source: Olist E-Commerce Public Dataset (Kaggle) · Period: Jan 2017 – Aug 2018

Steps:

1. Data Collection & Loading
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Data Visualization
5. Business Recommendations

Analysis Questions

Q1: Which sellers are most valuable by revenue & order volume?

Goal: Identify top contributors to platform revenue

Q2: How do monthly revenue & orders trend over time?

Goal: Understand platform growth patterns

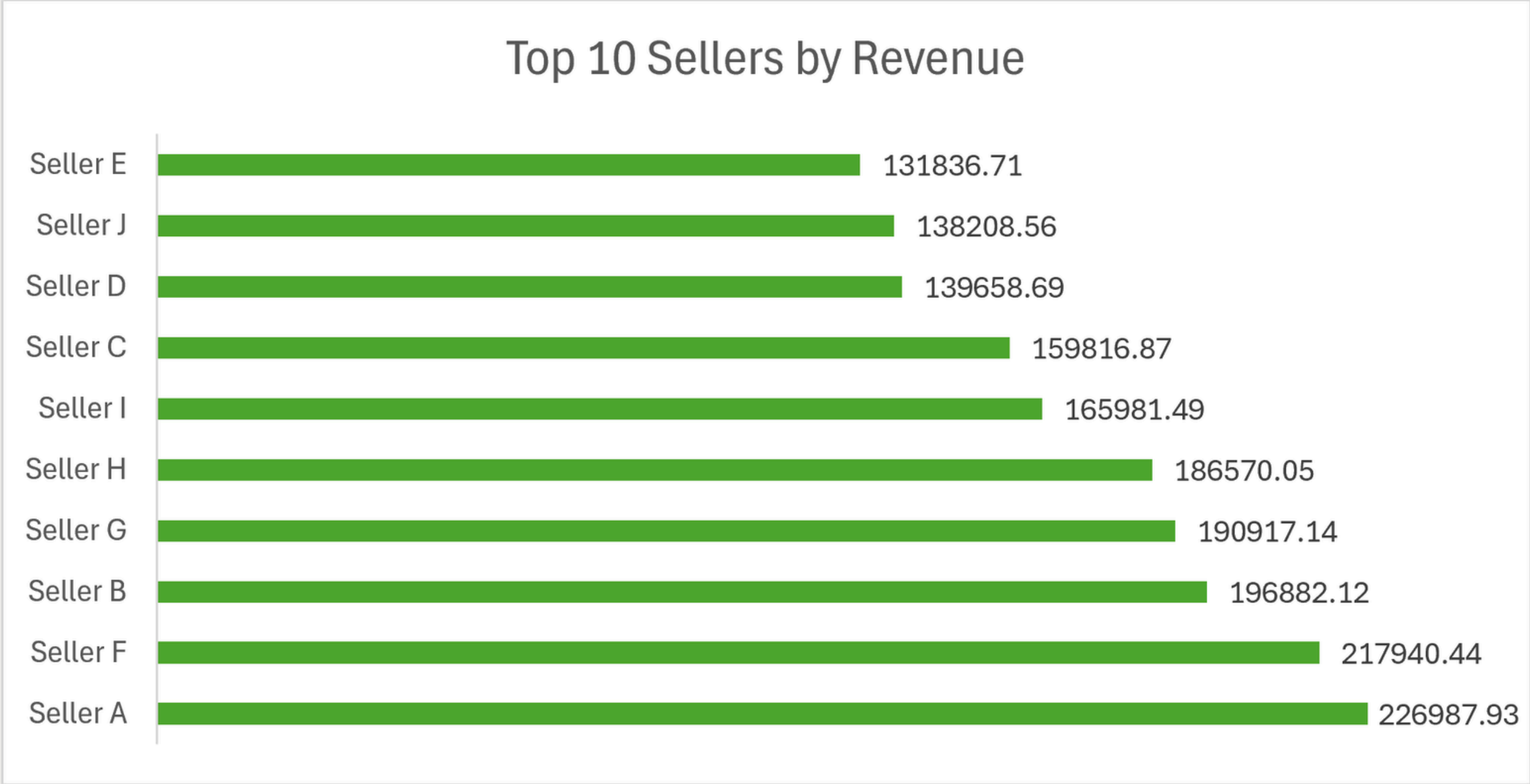
Q3: Which regions produce the best-performing sellers?

Goal: Determine priority areas for seller expansion and retention

Q4: Which product categories dominate platform sales?

Goal: Identify top categories for stock and promotion strateg

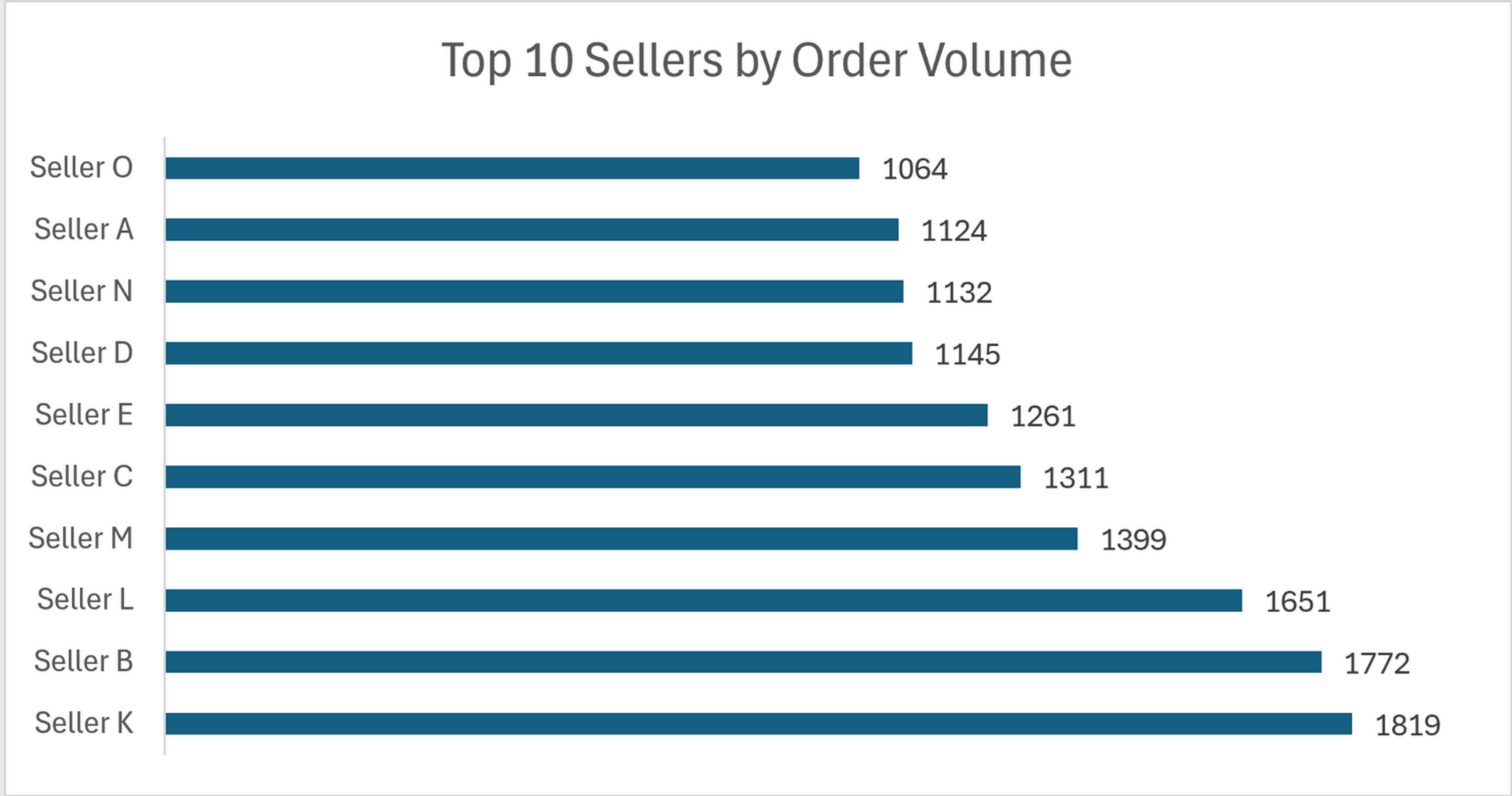
Top 10 Sellers by Revenue



Insight

Seller A leads with BRL 226,987.93 in revenue. The top 3 sellers contribute disproportionately more than ranks 4–10, revealing revenue concentration among a small group of top performers.

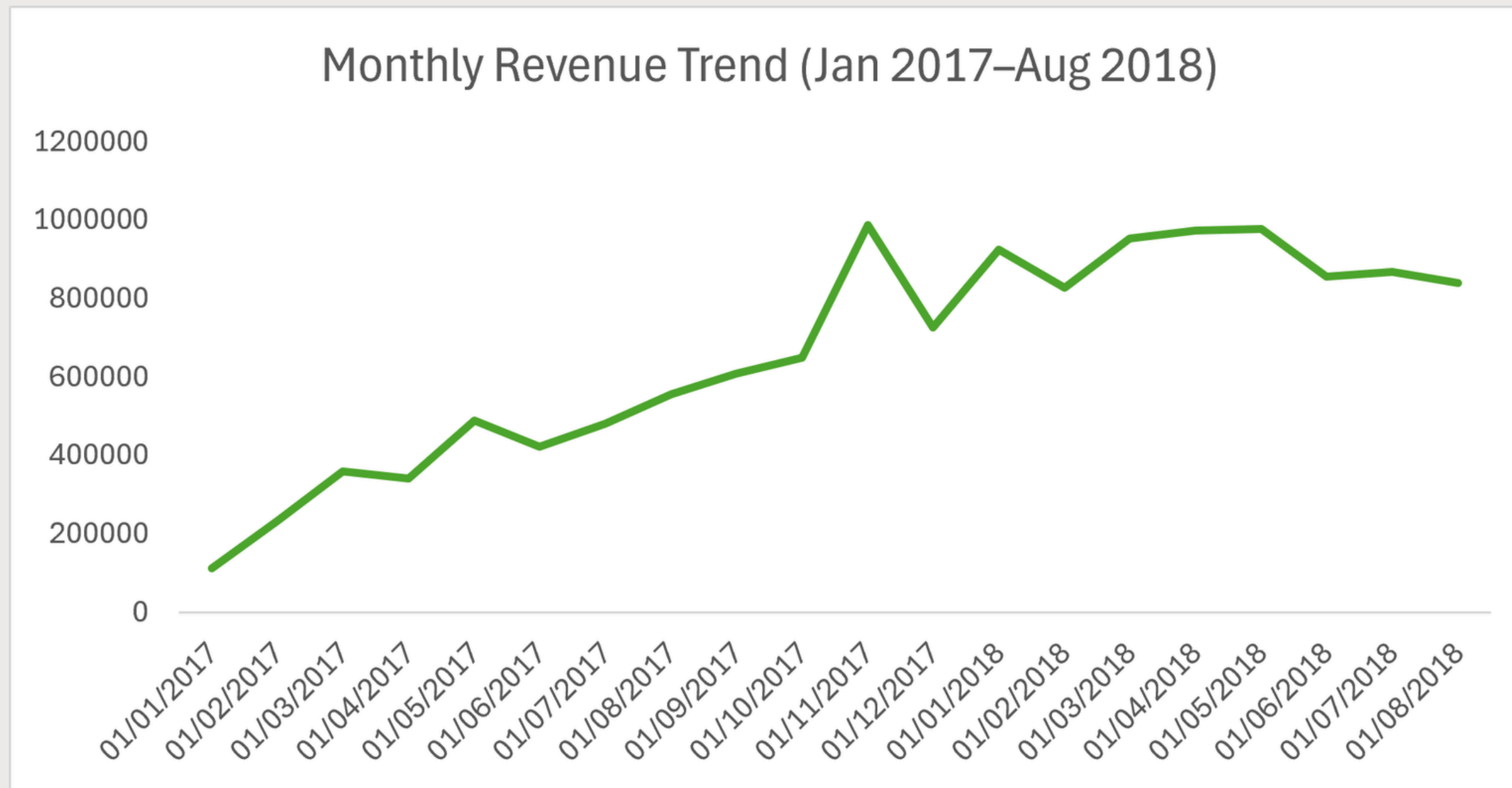
Top 10 Sellers by Order Volume



Insight

Most high-order sellers also appear among the top revenue earners, indicating that high order volume and high revenue frequently overlap rather than reflecting two separate seller strategies.

Monthly Revenue Trend



Insight

Revenue grew steadily from BRL 111,798.36 (Jan 2017) to a peak of BRL 987,765.37 (Nov 2017), then stabilized between BRL 826,437–977,545 throughout 2018 — reflecting Olist's rapid expansion phase followed by market maturity.

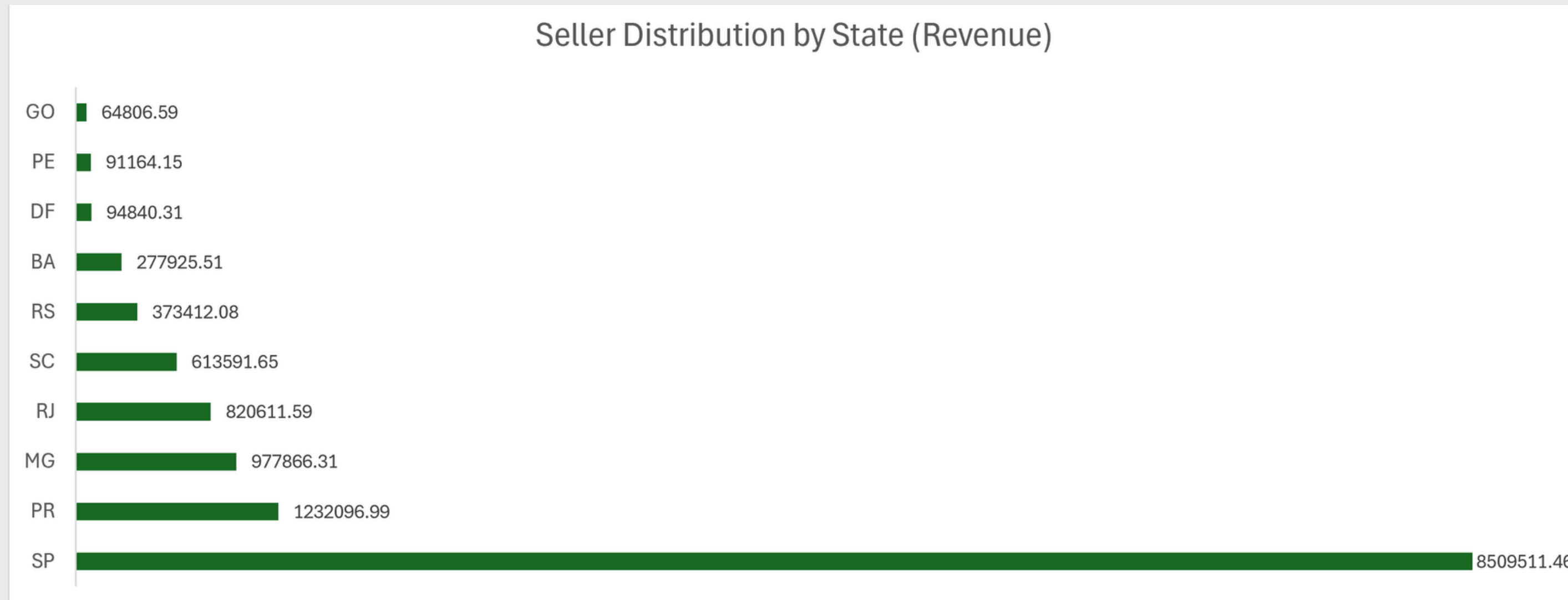
Monthly Order Volume Trend



Insight

Order volume peaked at 8,665 in Sep 2017 before stabilizing at 7,000–8,000 monthly. The parallel growth between volume and revenue confirms healthy platform scalability.

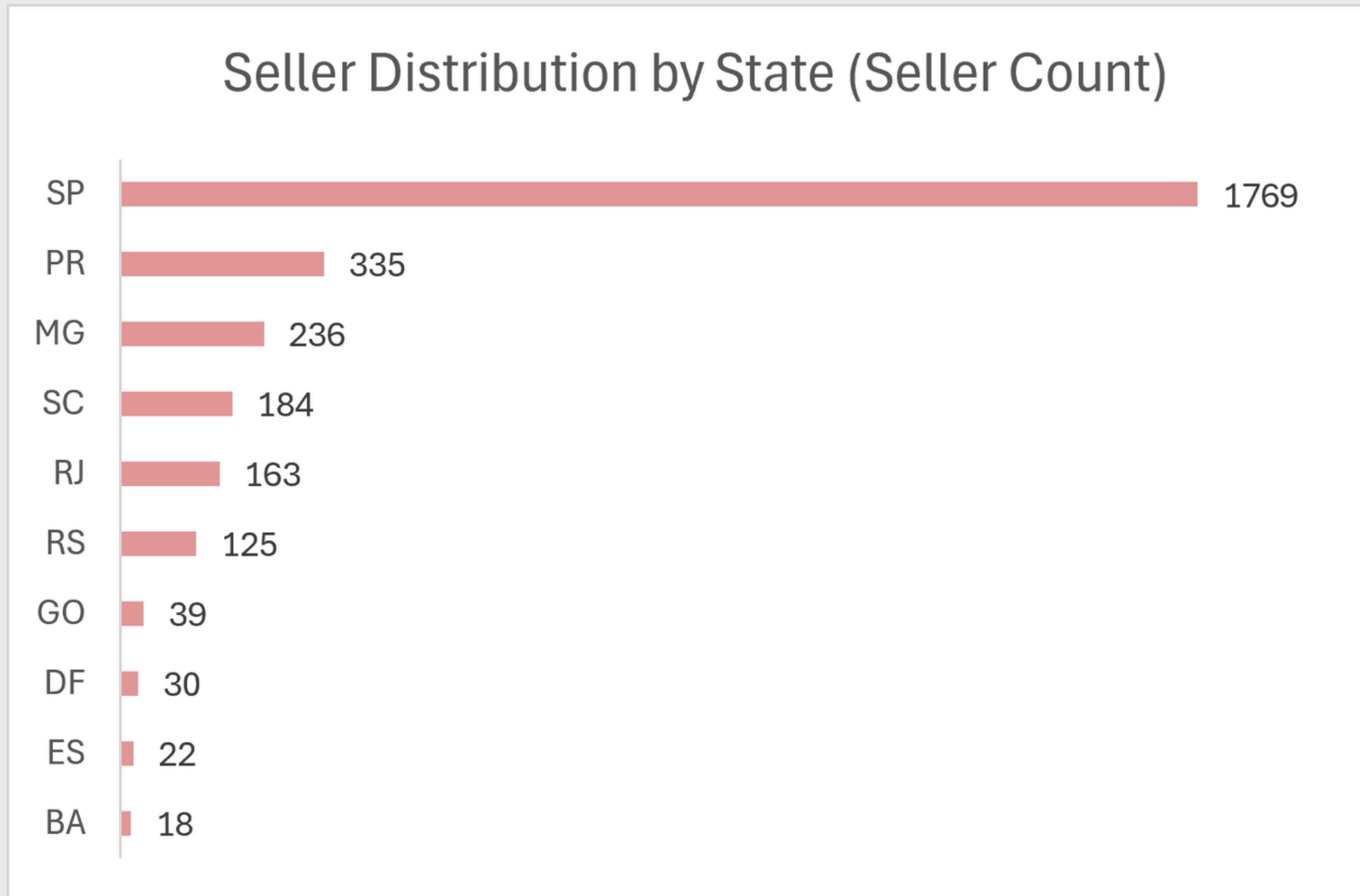
Seller Distribution by State (Revenue)



Insight

São Paulo dominates with BRL 8,509,511.46 — nearly 7x the second-ranked state (PR, BRL 1,232,096.99), making SP the undisputed priority market for seller acquisition and retention.

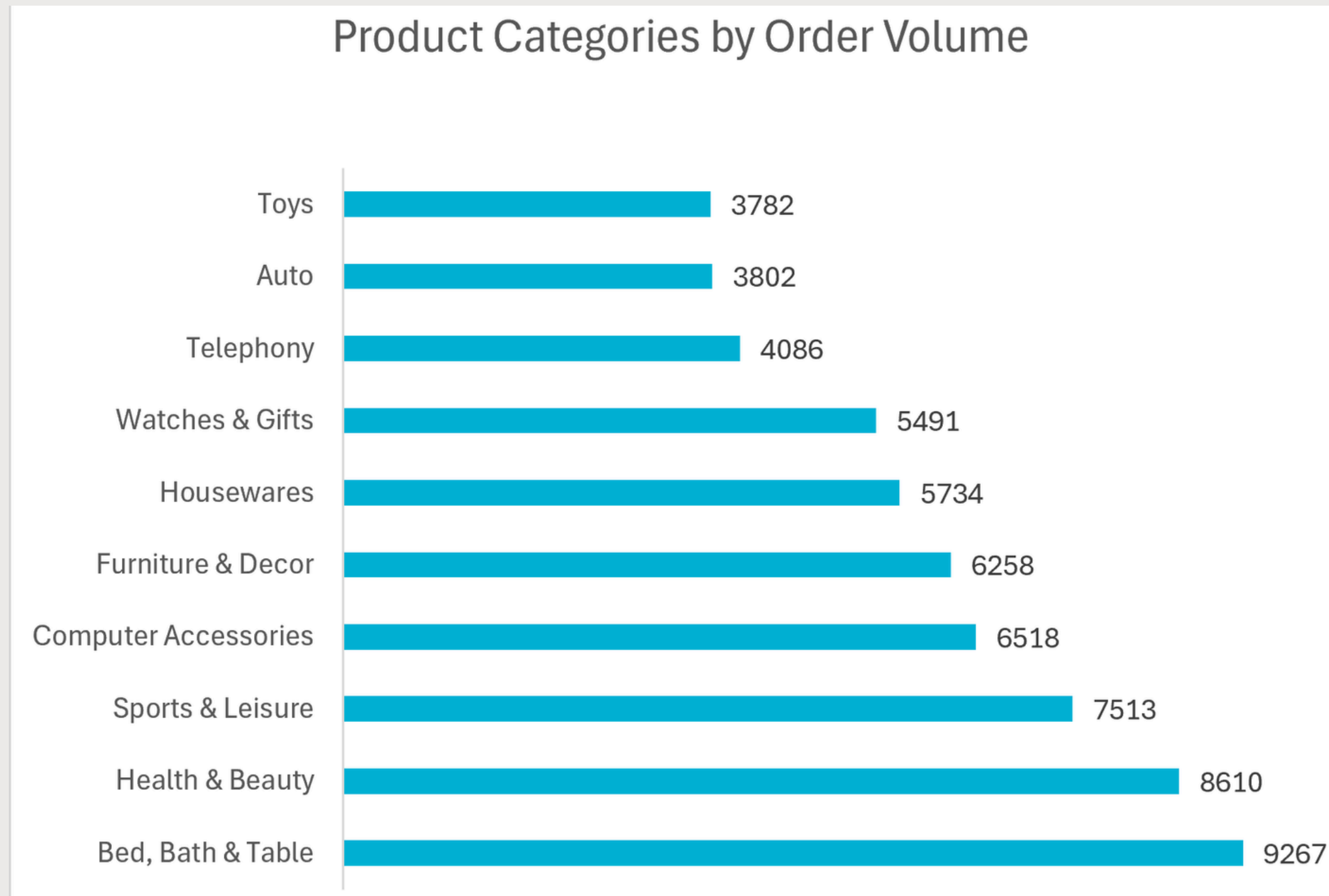
Seller Distribution by State (Seller Count)



Insight

São Paulo leads with 1,769 sellers. However, PR (335 sellers) generates more revenue per seller than MG (236 sellers) — PR shows stronger revenue efficiency despite fewer sellers, indicating MG has room for revenue optimization relative to its seller base.

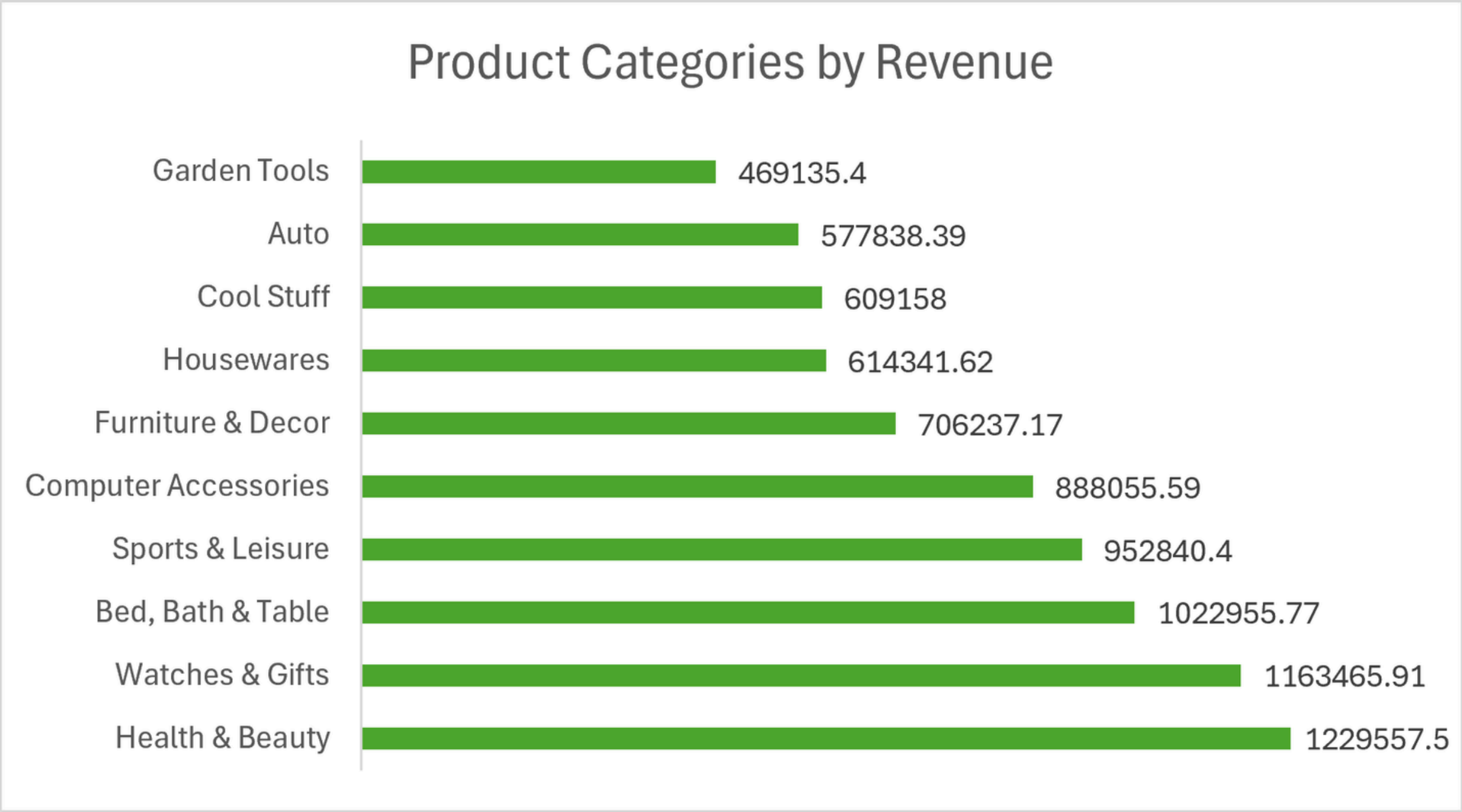
Product Categories by Order Volume



Insight

Bed, Bath & Table leads with 9,267 orders, followed by Health & Beauty (8,610) and Sports & Leisure (7,513) — these three categories represent the highest consumer demand on the platform.

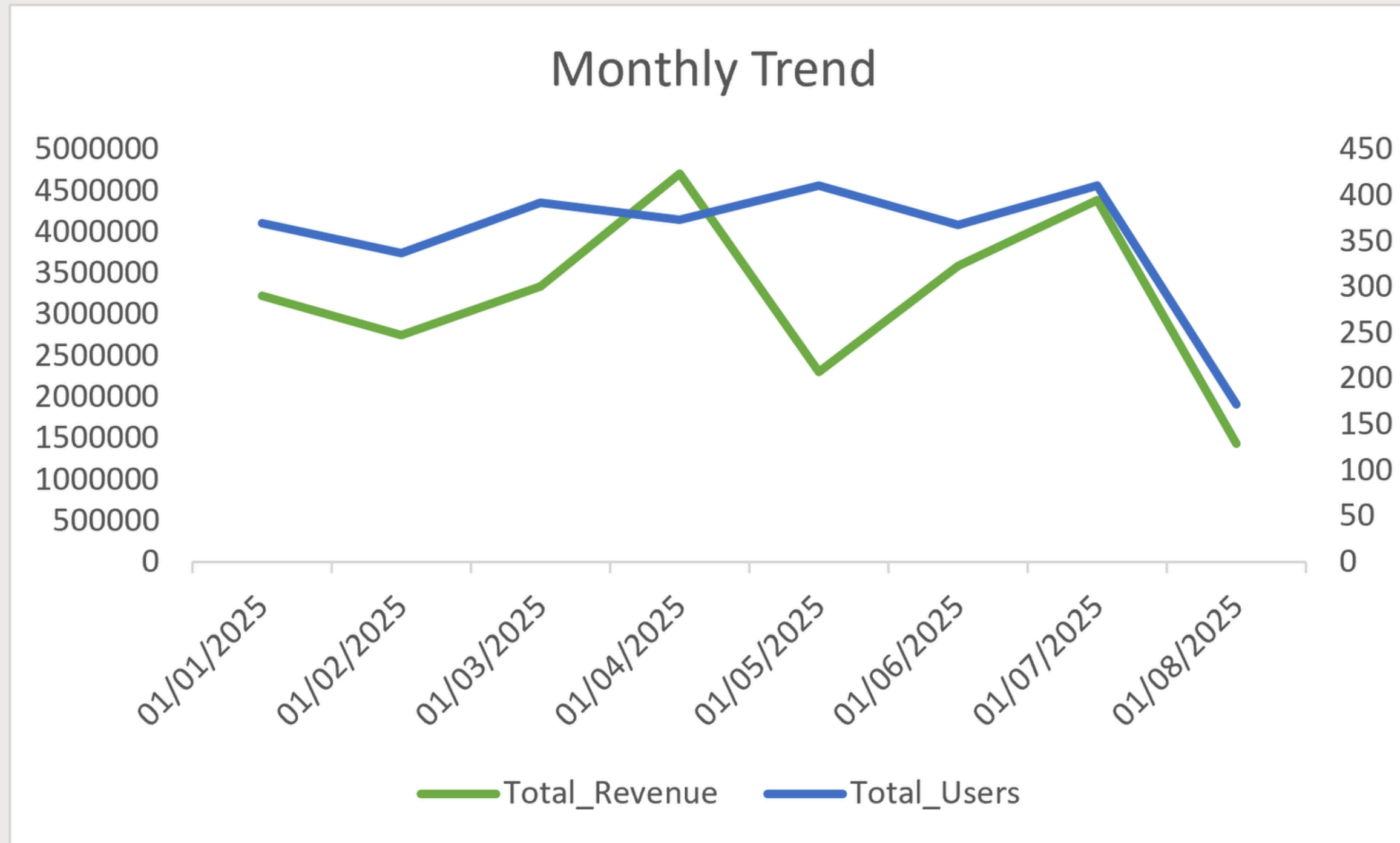
Product Categories by Revenue



Insight

Health & Beauty generates the highest revenue (BRL 1,229,557.50) despite ranking second in order volume — signaling higher average transaction value and effective pricing.

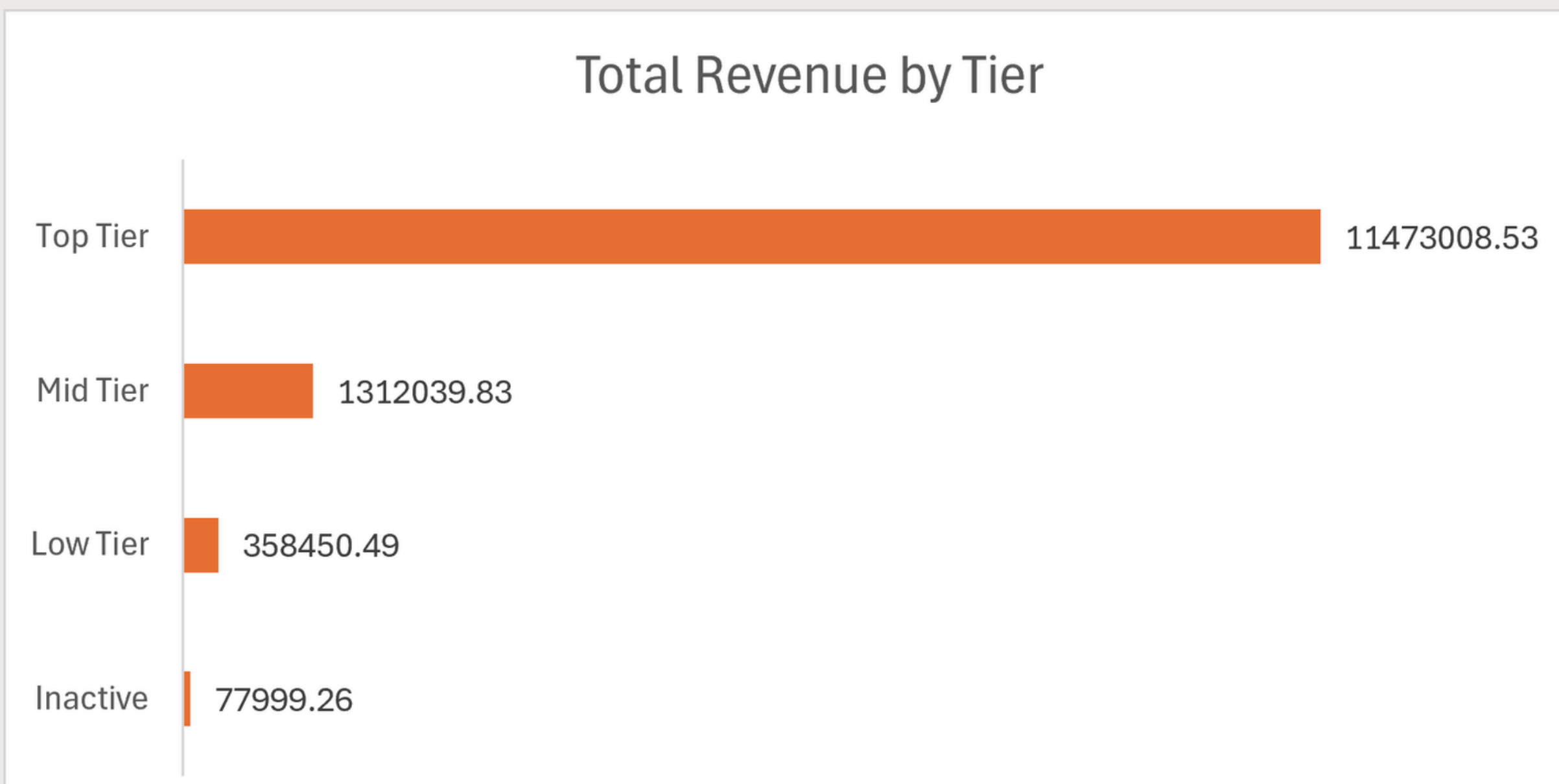
Total Revenue and Growth



Insight

Platform revenue grew consistently from BRL 111,798.36 (Jan 2017) to a peak of BRL 987,765.37 (Nov 2017), then stabilized between BRL 826,437–977,545 throughout 2018. Overall growth: 7.5x from January 2017 to August 2018.

Total Revenue by Tier



Insight

Top Tier sellers account for 86.8% of total platform revenue (BRL 11,473,008.53) despite representing only 25.0% of all sellers — a classic Pareto distribution signaling high dependency risk on a small seller base.

Seller Cohort Retention Heatmap

Sum of active_sellers		Column Labels																				
Row Labels		01/01/2017	01/02/2017	01/03/2017	01/04/2017	01/05/2017	01/06/2017	01/07/2017	01/08/2017	01/09/2017	01/10/2017	01/11/2017	01/12/2017	01/01/2018	01/02/2018	01/03/2018	01/04/2018	01/05/2018	01/06/2018	01/07/2018	01/08/2018	Grand Total
01/01/2017		219	166	162	139	149	136	125	136	123	126	124	109	115	108	109	102	88	87	80	88	2491
01/02/2017			236	150	134	141	111	122	121	107	107	101	107	91	89	88	93	74	78	72	65	2087
01/03/2017				164	99	93	87	81	82	77	71	82	64	67	58	57	54	52	52	45	48	1333
01/04/2017					116	65	51	61	56	53	44	52	47	46	37	41	38	36	30	33	28	834
01/05/2017						115	64	61	63	58	56	55	48	52	53	46	41	36	29	35	27	839
01/06/2017							70	30	28	33	23	27	27	24	18	21	21	17	15	17	16	387
01/07/2017								113	77	69	70	76	69	64	60	60	59	58	57	50	55	937
01/08/2017									120	66	57	69	55	57	62	55	50	49	49	40	42	771
01/09/2017										125	67	67	59	64	50	58	53	53	43	40	42	721
01/10/2017											141	97	79	83	67	71	69	67	56	61	48	839
01/11/2017												187	98	104	83	88	81	70	71	78	63	923
01/12/2017													84	51	46	46	41	37	33	31	31	400
01/01/2018														138	80	70	80	63	65	56	51	603
01/02/2018															117	63	61	53	50	57	50	451
01/03/2018																108	67	59	58	49	48	389
01/04/2018																	195	121	105	104	94	619
01/05/2018																		172	98	99	95	464
01/06/2018																			189	107	114	410
01/07/2018																				184	104	288
01/08/2018																					152	152
Grand Total		219	402	476	488	563	519	593	683	711	762	937	846	956	928	981	1105	1105	1165	1238	1261	15938

Insight

Sellers who joined in Jan–Mar 2017 show the strongest long-term retention, maintaining 50–80 active sellers even 12+ months after their first transaction, suggesting early cohorts are the platform's most loyal contributors.

Business Recommendations

Top Seller Retention Program

Create loyalty incentives for top 10 sellers: lower commission fees or homepage priority placement

1

2

Regional Expansion

Focus seller acquisition in MG and PR which show strong seller count but underoptimized revenue

Category Optimization

Prioritize stock and promotions for Health & Beauty and Bed, Bath & Table

3

Q4 Seasonal Campaign

Revenue peaks in Nov 2017 — schedule major promotions every Q4

4

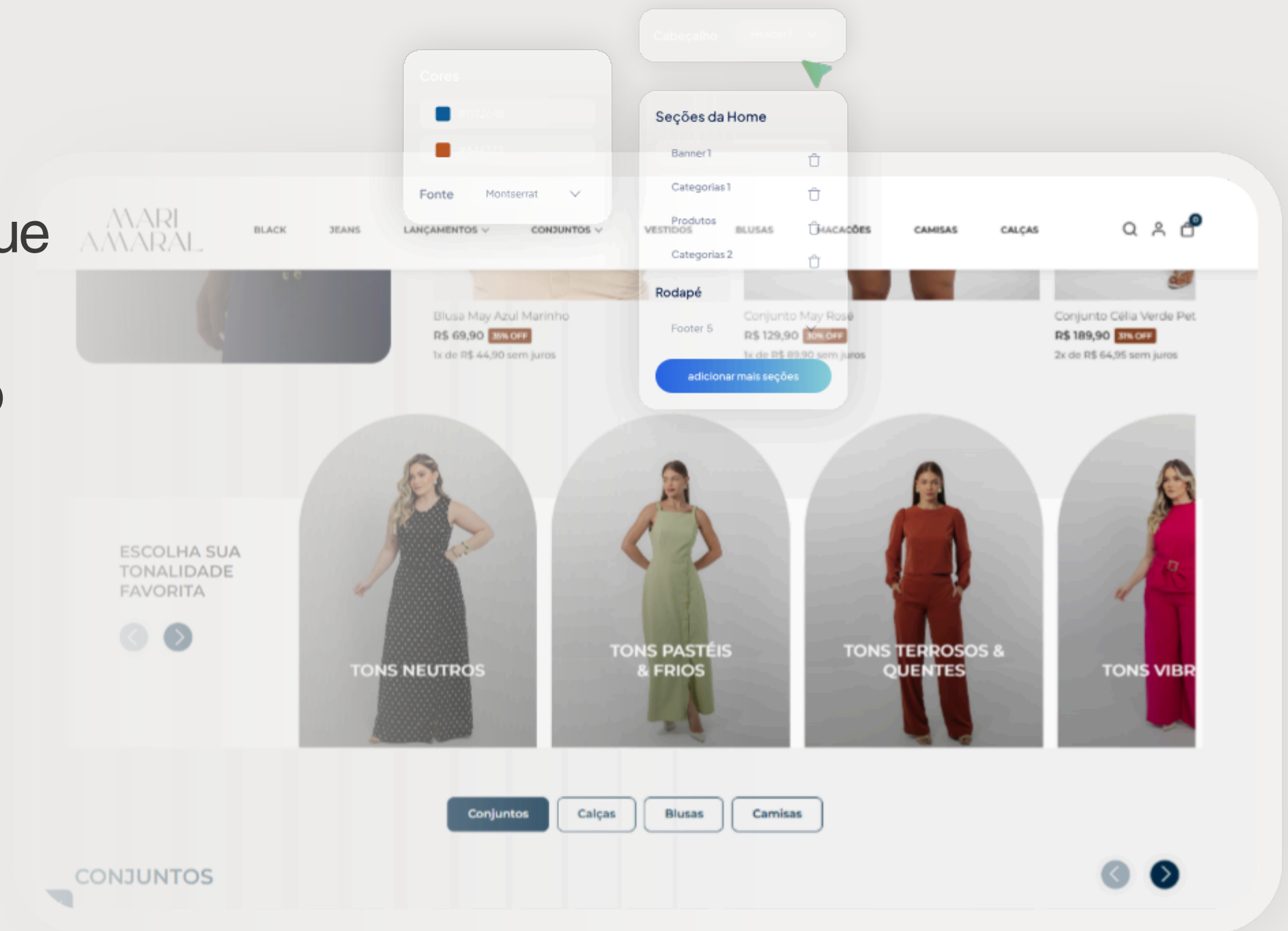
Seller Upselling Program

Push high-volume low-revenue sellers toward higher-value products through training

5

Conclusion

Based on the Olist dataset from January 2017 to August 2018, the platform demonstrated significant growth with revenue increasing 7.5x in 19 months. However, performance is heavily concentrated in São Paulo and a small group of elite sellers. Health & Beauty and Bed, Bath & Table emerged as the strongest categories. Olist has a strong business foundation with substantial growth potential through expansion into underoptimized regions like Minas Gerais and Paraná.



Thank You



Jessica Leo

[LinkedIn](#)



[Github](#)